Determination of column void volume using Gel Filtration Chromatography

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AIM:

To determine the column void volume V0 via gel filtration chromatography using Sephadex G-25 and Blue Dextran.

BACKGROUND :

Separation using Gel Filtration Chromatography is based on the separation of molecules based on their size and shape.

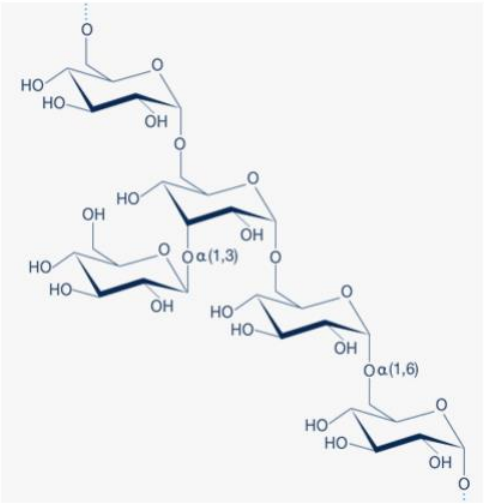
The setup consists of a stationary phase, which are formed by beads mode up of pores that span a relatively narrow size range in a long cylindrical column.

The mobile phase consists of a buffer, that is allowed to flow through the column.

The void volume, to be determined through this experiment, refers to the volume of empty space between the beads of the mobile phase, within the total volume of the column. Smaller molecules spend more time inside the beads than larger molecules. Therefore, the small molecules elute later, after a larger volume of mobile phase has passed through the column.

In summary, large molecules are excluded from the pores and travel through the column fastest, while small molecules can diffuse into the pores and elute later. The gels that are used as molecular sieves are cross-linked polymers. They are uncharged and inert, such that they do not bind or react with any materials being analysed.

Commonly used gels included Dextran, Polyacrylamide, and Agarose. For this experiment, Dextran (Sephadex) will be used, which is a homopolysaccharide of glucose residues. It shows various degrees of cross-linking to control pore size. The dry beads of Sephadex swell when water is added. It is mainly used for separation of small peptides and globular proteins with small to average molecular mass. Sephadex G-25 was used in this experiment, which separates molecules with molecular weights from 3000 to 70000. Molecules with molecular weights larger than 70000 will be excluded from the beads. Blue Dextran was used to measure the column void volume.



STRUCTURE OF SEPHADEX

PROCEDURE

A standard protocol was followed to perform this experiment:

1. A pre-refrigerated gel filtration column was clamped to the stand in a burette clamp. This was the primary apparatus for the experiment.

2. Blue Dextran (10 mg/ml) was taken in 50 mM Tris-HCl (pH-7.0) and 100 mM NaCl buffer. The volume of the Blue Dextran was ensured to be 1% to 3% of the column's bed volume.

3. The total content of Blue Dextran was mixed. The volume of the Blue Dextran was 2% of the bed volume (448 µL).

4. Buffer was added to the column until the level reached just at the top of the gel, while the stopcock was closed.

5. The Blue Dextran sample (2% of bed volume) was slowly pipetted into the column. The gel at the top of the column was not disturbed.

6. A tube was placed under the column. The stopcock was then opened and the buffer was allowed to move along the column with the sample. The buffer was drained into the 15 mL tube.

7. Buffer was continuously added to the top of the column to ensure that the gel did not run dry.

8. The stopcock was closed before the Blue Dextran sample escaped the column. The buffer runoff that was devoid of the sample was fully drained into the 15 mL tube, and the volume of the buffer collected was recorded.

9. The stopcock was opened again and the Blue Dextran sample was allowed to flow. The sample was collected in 1.5 mL vials, such that each vial collected only 500 µL of the blue colour elute. Buffer was slowly added to the top of the column to ensure that the gel did not run dry.

10. After the entire sample was collected into vials, the stopcock was closed.

11. 500 µL of buffer [50 mM Tris-HCl (pH-7.0) and 100 mM NaCl] was added to each of the 1.5 mL vials containing the sample, and mixed well.

12. The absorbance of these samples was recorded at 610 nm wavelength using a Spectrophotometer. A blank containing only buffer was used.

13. The vial number showing the maximum absorbance value was recorded.

14. The volume of the buffer collected in the tube, and the volume of the Blue Dextran collected in 1.5 mL vials up to the vial with the maximum absorbance were added. The total volume obtained is the V_0 or the void volume of the column

DATA ANALYSIS

The total buffer volume V_{buffer} collected in the tube was recorded.

$$V_{buffer} = 16.5 \text{ mL}$$

The data obtained from the spectrophotometry of the vials containing Blue Dextran is tabulated below :

VIAL NO.	ELUTION VOLUME	ABSORBANCE AT 610 NM
1	17	0.062
2	17.5	0.11
3	18	0.16
4	18.5	0.205
5	19	0.293
6	19.5	0.329
7	20	0.359
8	20.5	0.364
9	21	0.368
10	21.5	0.377
11	22	0.379
12	22.5	0.38
13	23	0.383
14	23.5	0.385
15	24	0.418
16	24.5	0.402
17	25	0.399
18	25.5	0.392

19	26	0.331
20	26.5	0.319
21	27	0.257
22	27.5	0.239
23	28	0.232
24	28.5	0.218
25	29	0.179
26	29.5	0.146
27	30	0.119
28	30.5	0.097
29	31	0.094
30	31.5	0.062
31	32	0.027
32	32.5	0.022
33	33	0.02
34	33.5	0.007

Table 1

The elution volume in Table 1 refers to the successive volumes in the vials after the 16.5 mL buffer (V_{buffer}) was collected. Hence, the readings start from 16.5 mL. Since each vial contains 0.5 mL (500 µL) of Blue Dextran, the values increase by 0.5 mL in each vial.

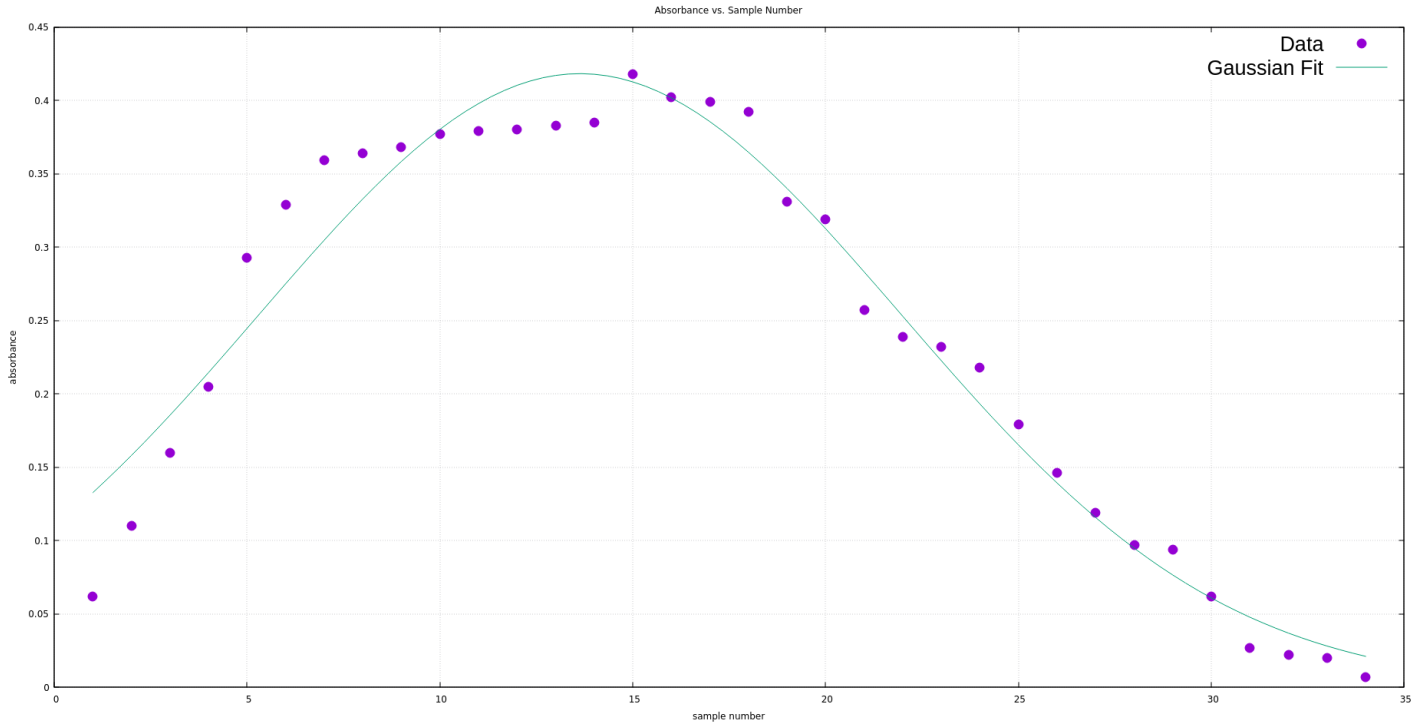
• It can be concluded from the spectrophotometry data that vial 15 pertains to the maximum absorbance value of 0.418, which has an elution volume of 24.0 mL. This is the void volume V_0 of the gel.

$$V_0 = 16.5 \text{ mL buffer} + 15\times 0.5 \text{ mL Blue Dextran} = 24 \text{ mL}.$$

CONCLUSION:

Gel Filtration Chromatography was conducted in the laboratory using Sephadex G-25 and Blue Dextran, and Column void volume was calculated using spectrophotometric analysis. In principle, the particles of Blue Dextran are larger in size than the pore size of the gel, allowing it to flow freely and elute through the column. Thus, it can be used to

calculation the void volume. The column void volume, V_0 was recorded to be 24 mL.



Absorbance vs sample number graph